I E S

COLLEGE OF ENGINEERING

CHITTILAPPILLY, THRISSUR.

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

ANTENNA EXPERIMENTS

Name:……...

Branch:…… ……..Semester:…….:……

UniversityRegisterNo:……...

Lab In charge

Experiment No: Date:

FAMILIARIZATION OF ANTENNA SIMULATION SOFTWARE: INTRODUCTION TO HFSS

Aim:

To familiarize with antenna simulation software HFSS.

Equipment Required:

- Computer
- HFSS Software

Theory:

The name HFSS stands for High Frequency Structure Simulator. HFSS is a high- performance full-wave electromagnetic (EM) field simulator for arbitrary 3D volumetric passive device modeling that takes advantage of the familiar Microsoft Windows graphical user interface. It integrates simulation, visualization, solid modeling, and automation in an easy-to-learn environment where solutions to your 3D EM problems are quickly and accurately obtained. Ansoft HFSS employs the Finite Element Method (FEM), adaptive meshing, and brilliant graphics to give you unparalleled performance and insight to all of your 3D EM problems. Ansoft HFSS can be used to calculate parameters such as S Parameters, Resonant Frequency, and Fields.

Ansoft HFSS Online Help provides assistance while you are working:

- To get help about a specific, active dialog box, click the Help button in the dialog box orpresstheF1 key.
- Select them enuitem Help>Contents to access the online help system.
- Tooltips are available to provide information about tools on the toolbars or dialog boxes. When you hold the pointer over a tool for a brief time, a tooltip appears to display the name of the tool.
- As you move the pointer over a tool or click a menu item, the Status Bar at the bottom of the Ansoft HFSS window provides a brief description of the function of the tool or menu item.
- The Ansoft HFSS Getting Started guide provides detailed information about using HFSS to create and solve3D EM projects.

Procedure:

Ansoft Terms

The Ansoft HFSS window has several optional panels:

- Project Manager
- Message Manager
- Property Window
- Progress Window
- 3DModelerWindow

These above managers and windows are shown in Fig.1 and their details are given in coming sections.

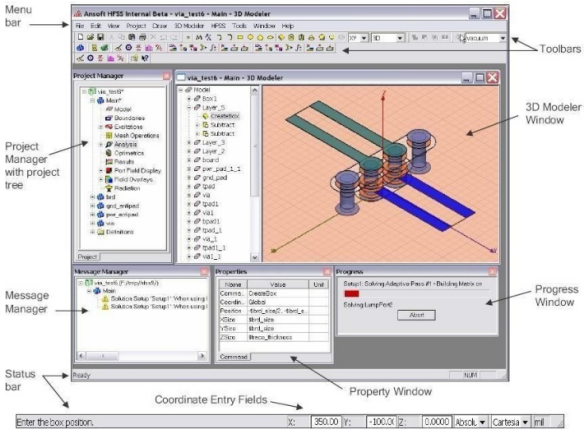


Fig.1:Ansoft HFSS Window

Project Manager

A Project Manager which contains a design tree which lists the structure of the project is shown in Fig.2.

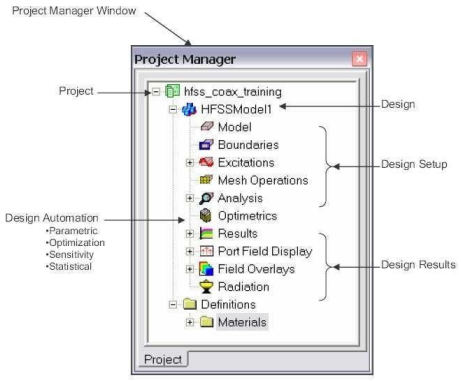


Fig.2:Ansoft HFSS Project Manager.

Message Manager :

A Message Manager that allows you to view any errors or warnings that occur before you begin a simulation is shown in Fig.3.



Fig.3: Ansoft HFSS Message Manager

Property Window

A Property Window that displays and allows you to change model parameters or attributes is shown in Fig.4.

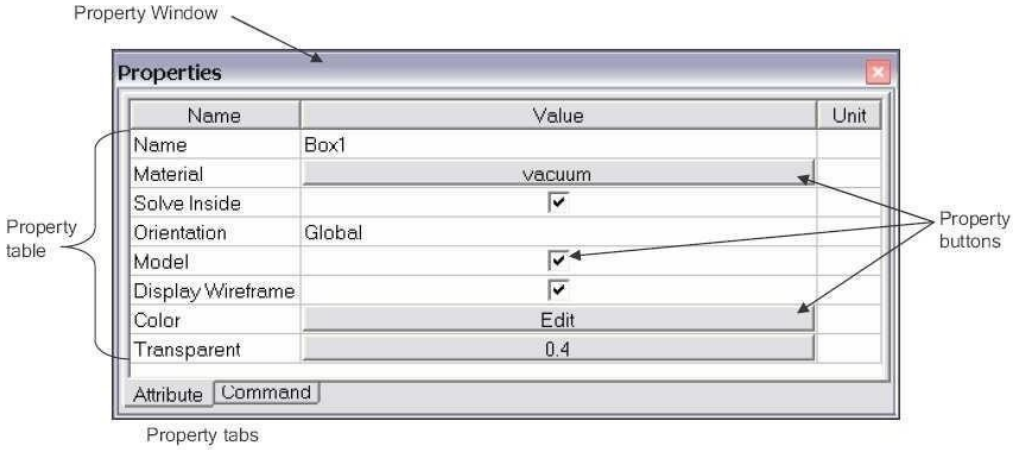


Fig.4:AnsoftHFSS Property Window.

Progress Window

A Progress Window that displays solution progress is shown in Fig.5.



Fig.5:Ansoft HFSS Progress Window.

In the Ansoft HFSS Desktop, each project can have multiple designs and each design is displayed in a separate window. You can have multiple projects and design windows open at the same time. Also, you can have multiple views of the same design visible at the same time.

To arrange the windows, you can drag them by the title bar, and resize them by dragging a corner or border. Also, you can select one of the following menu options: Window >Cascade, Window >Tile Vertically or Window >Tile Horizontally.

To organize your Ansoft HFSS window,you can iconize open designs. Click the Iconize**symbol in the upper right corner of the document border. An icon appears in the lower part of the Ansoft HFSS window. If the icon is not visible, it may be behind an other open document. Resize any open documents as necessary. Select the menu item Window > Arrange Icons to arrange them at the bottom of the Ansoft HFSS window. Select the menu item Window > Close All to close all open design. You are prompted to save unsaved designs.

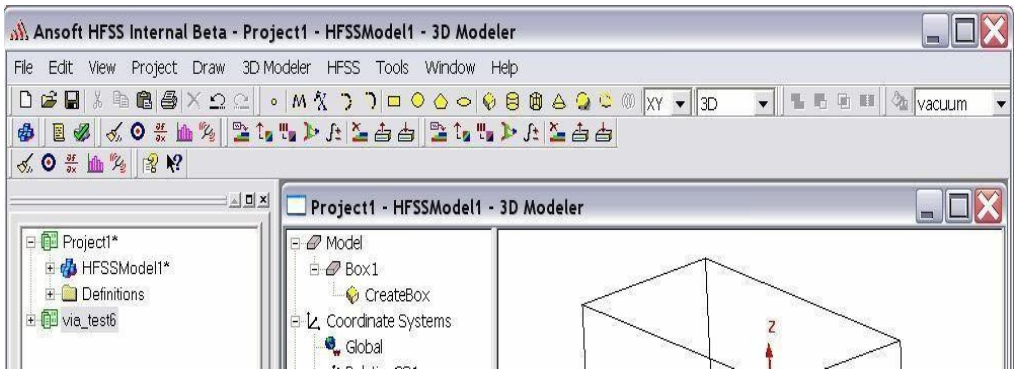


Fig.6: DesignWindow.

Toolbars:

The toolbar buttons are shortcuts for frequently used commands. Most of the available toolbars are displayed in this illustration of the Ansoft HFSS initial screen, but your Ansoft HFSS window probably will not be arranged this way.

You can customize your tool bar display in a way that is convenient for you. Some tool bars a real ways displayed; other toolbars display automatically when you select a document of the related type. For example, when you select a 2D report from the project tree, the 2D report toolbar displays, as shown in Fig.7.

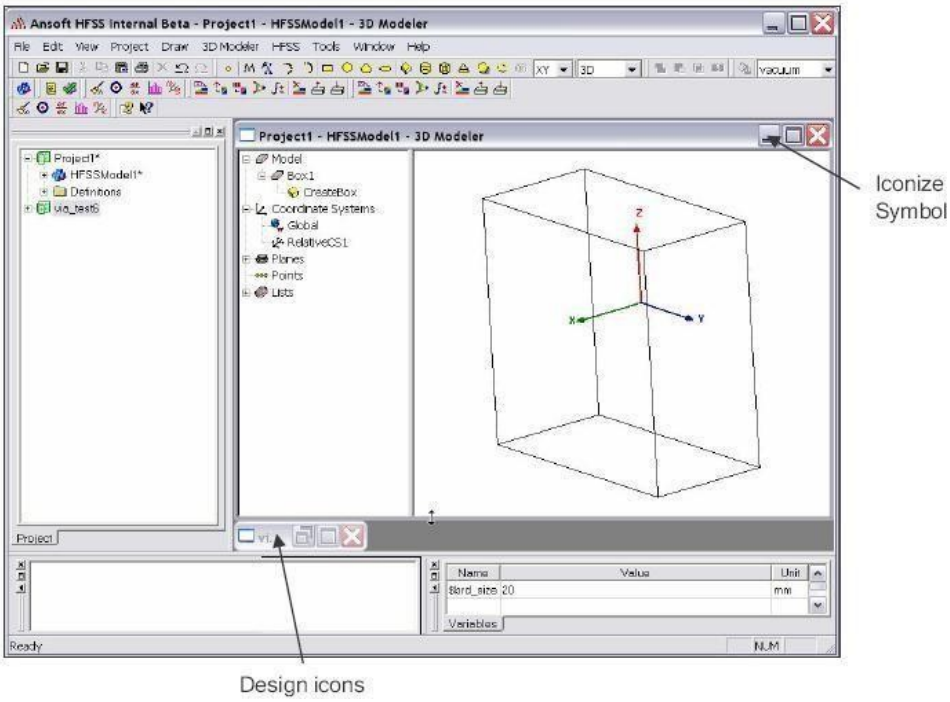


Fig.7:Ansoft HFSS Toolbars.

To display or hide individual toolbars

- Right-click the Ansoft HFSS window frame.

  A list of all the toolbars is displayed. The toolbars with a check mark beside them are visible; the toolbars without a check mark are hidden. Click the toolbar name to turn its display on or off.

- To make changes to the toolbars, select the menu item Tools > Customize.

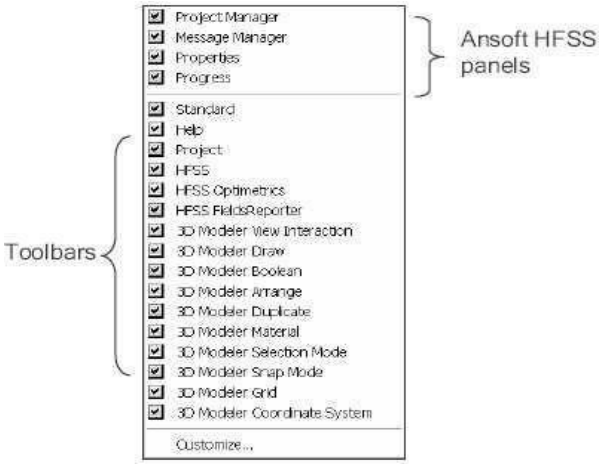


Fig.8:AnsoftHFSS Panels and Toolbars.

Ansoft HFSS Desktop

The Ansoft HFSS Desktop provides an intuitive, easy-to-use interface for developing passive RF device models. Creating designs, involves the following:

- Parametric Model Generation –creating the geometry, boundaries and excitations.
- Analysis Setup–defining solution setup and frequency sweeps.
- Results–creating2Dreportsandfieldplots.
- Solve Loop –the solution process is fully automated.

To understand how these processes co-exist, examine the illustration shownInFig.9.

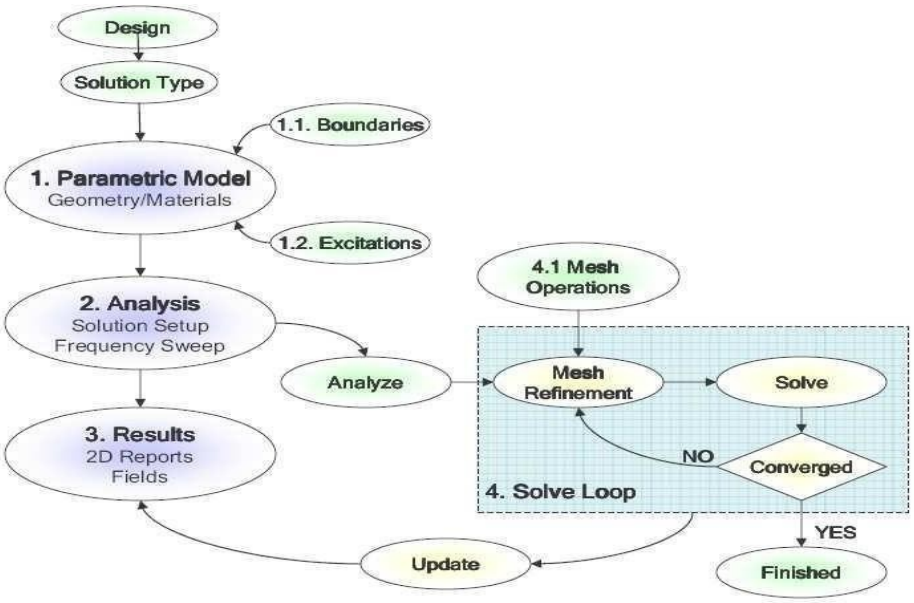


Fig.9:AnsoftHFSS Desktop.

Opening a HFSS project

This section describes how to open an existing project.

Opening a new project:

- In an Ansoft HFSS window, select the menu item File>New.
- Select the menu Project > Insert HFSS Design.

Opening an Existing HFSS project:

- In an Ansoft HFSS window, select the menu File > Open. Use the Open dialog to select the project.
- Click Open to open the project.

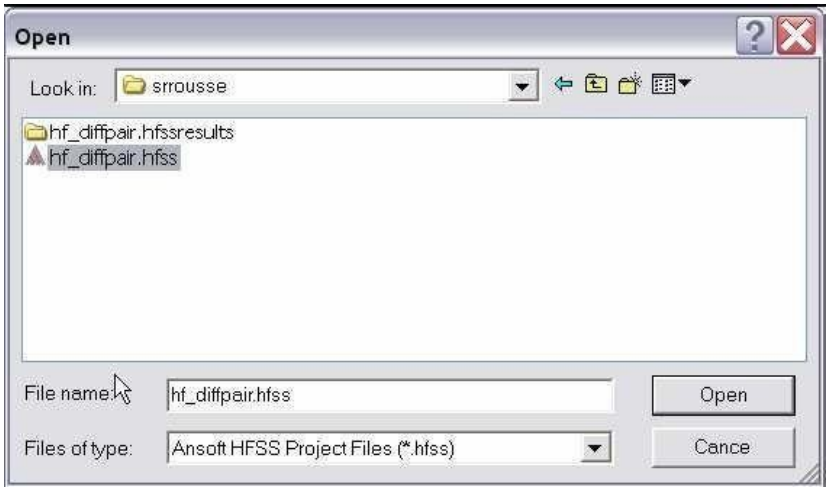


Fig.10: OpeningaHFSSproject.

Set Solution Type:

This section describe show to set the Solution Type. The Solution Type defines the type of results, how the excitations are defined, and the convergence. The following Solution Types are available:

- Driven Modal - calculates the modal-based S-parameters. The S-matrix solutions will be expressed in terms of the incident and reflected powers of waveguide modes.
- Driven Terminal - calculates the terminal-based S-parameters of multi conductor transmission line ports. The S-matrix solutions will be expressed in terms of terminal voltages and currents.
- Eigen mode – calculate the Eigen modes, or resonances, of a structure. The Eigen mode solver finds the resonant frequencies of the structure and the fields at those resonant frequencies.

Convergence :

- Driven Modal – Delta S for modal S-Parameters. This was the only convergence method available for Driven Solutions in previous versions.
- Driven Terminal New–Delta S for the single-ended or differential nodal S-Parameters.
- Eigen mode - Delta F

To set the solution type:

Select the menu item HFSS > Solution Type Solution Type Window:

Choose one of the following:

  - Driven Modal
  - Driven Terminal
  - Eigen mode

- Click the OK button

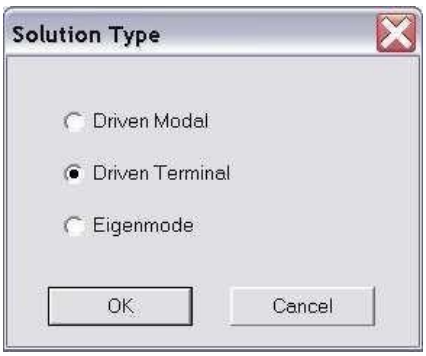


Fig.11:Solution Type.

The following Fig.12 shows the3D Modeler window.

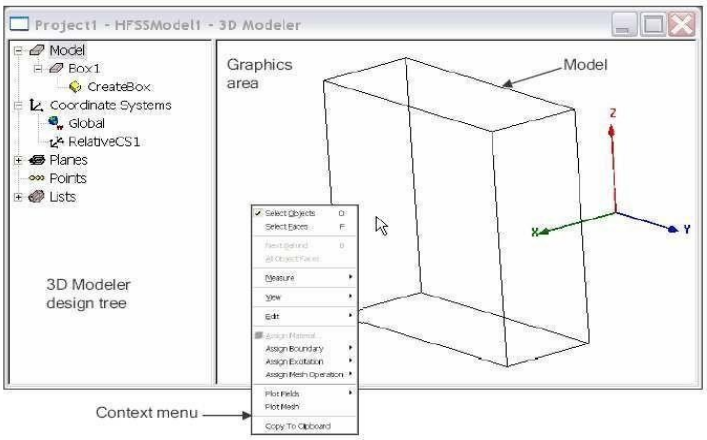


Fig.12: 3DModeler

- 3D Modeler Design Tree – The 3D Modeler Design Tree is an essential part of the user interface. From here you may access the structural elements in addition to any object dependencies and attributes.
- Context Menus – Context menus are a flexible way of accessing frequently used menu commands for the current context. The contents of these menus change dynamically and are available throughout the interface by clicking the right mouse button.
- Graphics Area–The graphics area is used to interact with the structural elements.

When using the 3D Modeler interface, you will also interact with two additional Interfaces:

- Property Window – The Property Window is used to view or modify the attributes and dimensions of structural objects is shown in Fig.13.
- Status Bar/Coordinate Entry – The Status Bar on the Ansoft HFSS Desktop Window displays

the Coordinate Entry fields that can be used to define points or offsets during the creation of structural objects is shown in Fig.14.

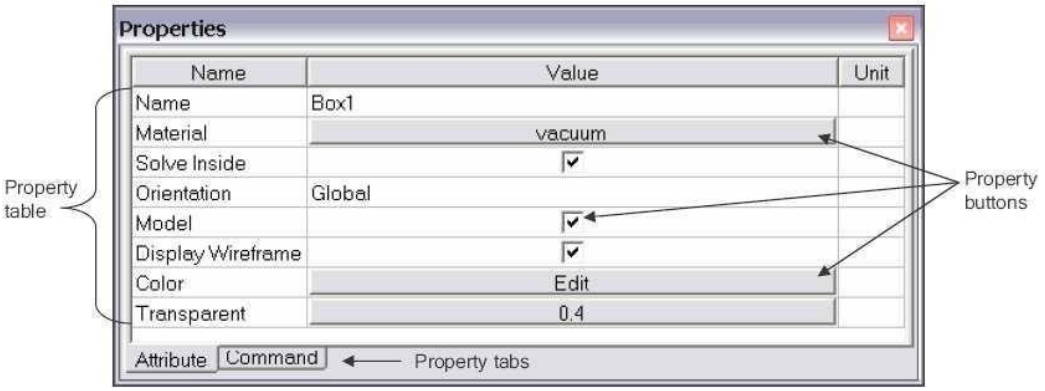


Fig.13:HFSS Property window.

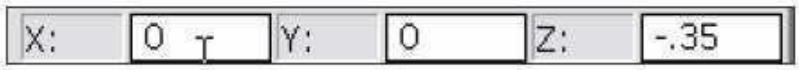


Fig.14:StatusBar/Coordinate Entry.

Grid Plane

To simplify the creation of structural primitives, a grid or drawing plane is used. The drawing plane does not in any way limit the user to two dimensional coordinates but instead is used as a guide to simplify the creation of structural primitives. The drawing plane is represented by the active grid plane.(The grid does not have to be visible). To demonstrate how drawing planes are used, review the following section:

- Creating and Viewing a Simple Structure

Creating3Dstructuralobjects is accomplished by performing the following steps:

- Set the grid plane
- Create the base shape of the object
- Set the Height

Create a Box

We will investigate creating a box to demonstrate these steps. These steps assume that project and a HFSS design have already been created. Three points are required to create the box. The first two form the base rectangle and the third sets the height.

Point 1: Defines the start point of the base rectangle

Point2: Defines the size of the base rectangle

Point3: Defines the height of the Box

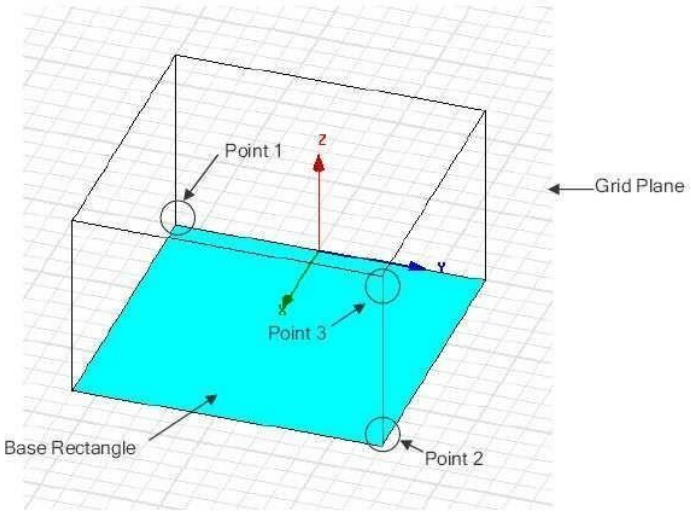
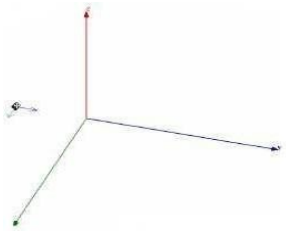


Fig.15:Creating a Box using Status Bar/Coordinate Entry.

- Select the menu item3DModeler>Grid Plane>XY
- Use the mouse to create the base shape.



- Set the start point by positioning the active cursor and click the left mouse button.
- Position the active cursor and click the left mouse button to set the second point that forms the base rectangle.

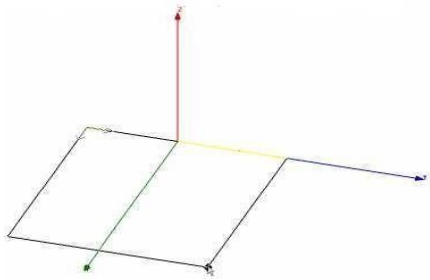


Fig.16:Grid

User Defined Project Material :

To define custom materials, click the Add Material button from the material definition window. The following dialog will appear. Enter the material definitions and click the OK.

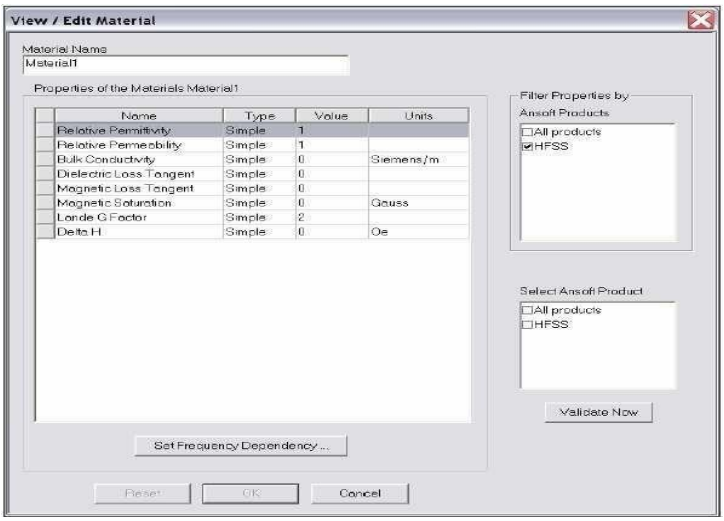


Fig.17:User Defined Project Material.

Changing the View:

You can change the view at any time (even during shape generation) by using the following commands:

Toolbar:

- Rotate– The structure will be rotated around the coordinate system.
- Pan–The structure will be translated in the graphical area.
- Dynamic Zoom – Moving the mouse upwards will increase the zoom factor while moving the mouse downwards will decrease the zoom factor.
- Zoom In/Out – In this mode a rubber band rectangle will be defined by dragging the mouse. After releasing the mouse button, the zoom factor will be applied.

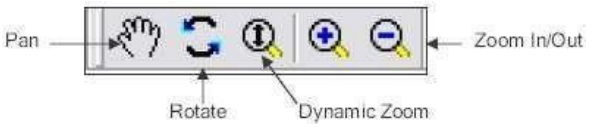


Fig.18:Context Menu

Right click in the graphics area and select the menu item View and choose from the options outlined in the Toolbar section. The context menu also offers the following:

- Fit All–This will zoom the defined structure to a point where it fits in the drawing area.
- Fit Selection–These fits only the selected objects into the drawing area.
- Spin– Drag the mouse and release the mouse button
- To start the object spinning. The speed of the dragging prior to releasing the mouse controls the speed of the spin.

- Animate – Create or display the animation of parametric geometry.

Shortcuts:

Since changing the view is a frequently used operation, some useful shortcut keys exist. Press the appropriate keys and drag the mouse with the left button pressed:

- ALT+Drag– Rotate
- Shift+Drag–Pan
- ALT + Shift + Drag – Dynamic Zoom

Shortcuts –Pre defined Views

Visibility:

The visibility of objects, Boundaries, Excitations ,and Field Reports can be controlled from the menu item View>Visibility.

Hide Selection:

The visibility of selected objects can be set hidden by selecting the object(s) and choosing the menu View>Hide Selection >All Views.

Background Color:
To set the background color, select the menu item View> Modify Attributes>Background Color.

Addition View Settings:
Additional attributes of the view such as the projection, orientation, and lighting can be set from the menu item View >Modify Attributes.

Applying Structural Transformations:

So far, we have investigated hot to model simple shapes and how to change the view of the model. To create more complicated models or reduce the number of objects that need to be created manually we can apply various transformations. The following examples assume that you have already selected the object(s) that you wish to apply a transformation. You can select the transformation options from the menu item Edit >

- Arrange>
  - Move–Translates the structure along a vector.
  - Rotate– Rotates the shape around a coordinate axis by an angle.
  - Mirror– Mirrors the shape around a specified plane.
  - Offset–Performs a uniform scale in x,y,and z.
- Duplicate>
  - Along Lines–Create multiple copies of an object along a vector.
  - Around Axis – Create multiple copies of an object rotated by a fixed angle around the x,y, or z axis.
  - Mirror-Mirrors the shape around a specified plane and creates Duplicate.

- Scale– Allows non-uniform scaling in the x,y, or z direction

The faces of an object can also be moved to alter the shape of an existing object. To move the faces of an object, select the menu item 3D Modeler > Surfaces > Move Faces and select Along Normal or along Vector.

Combine Objects by Using Boolean Operations:

Most complex structures can be reduced to combinations of simple primitives. Even the solid primitives can be reduced to simple 2D primitives that are swept along a vector or around an axis (Box is a square that is swept along a vector to give it thickness). The solid modeler supports the following Boolean operations:

- Unite–combine multiple primitives, United is joint objects.
- Subtract–remove part of a primitive from another.
- Split –break primitives into multiple parts.
- Intersect–keep only the parts of primitives that overlap.
- Sweep – turn a 2D primitive into a solid by sweeping: Along a Vector, Around an Axis, along a Path.
- Connect – connect 2D primitives. Use Cover Surfaces to turn the connected object into a solid.
- Section–generate2Dcross-sections ofa3Dobject.

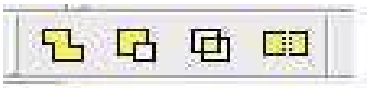


Fig.19:Unit, Subtract, Intersect, Split

Result:

Experiment No : 10

Date:

Dipole Antenna

Aim

Design and Simulate a Dipole Antenna of resonant frequency, f_r=700MHz.

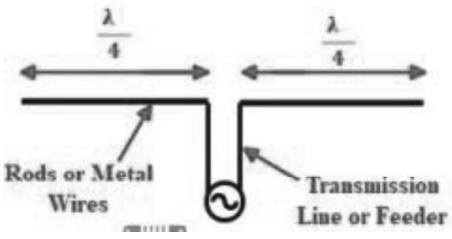
Theory

A dipole antenna (also known as a doublet or dipole aerial) is defined as a type of RF (Radio Frequency) antenna, consisting of two conductive elements such as rods or wires. The dipole is any one of the varieties of antenna that produce a radiation pattern approximating that of an elementary electric dipole. Dipole antennas are the simplest and most widely used type of antenna.

A 'dipole' means 'two poles' hence the dipole antenna consists of two identical conductive elements such as rods or metal wires. The length of the metal wires is approximately half of the maximum wavelength (i.e.,λ/2) in free space at the frequency of operation.

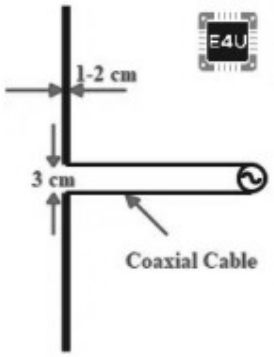
This wire or rod is split at the centre, and the two sections are separated by an insulator, these sections are known as an antenna section.

These two antenna sections are connected to a feeder or coaxial cable at the end closest to the centre of the antenna. Note that wavelength is the distance between two consecutive maximum or minimum points. The basic dipole antenna with the centre feed point is shown in the figure below.



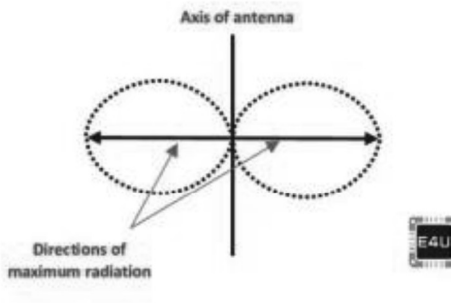
FM Dipole Antenna

FM dipole antenna is defined as the vertically polarized half-wave half dipole antenna. The FM dipole antenna is mostly used as it is to build and it is suitable to provide improved reception of VHF FM broadcast. The FM dipole antenna is shown in the figure below.



FM Dipole Antenna

The radiation pattern of the FM dipole antenna is perpendicular to the axis of the antenna i.e., It is horizontal because the FM dipole antenna is a vertically polarized half-wave dipole antenna. The radiation pattern of the FM dipole antenna is shown in the below figure.



Radiation Pattern of a FM Dipole Antenna

Procedure

1. Creating the project
2. Draw the cylinder(dipole arm). Right click on the **CreateCylinder** to change the dimensions and position of the cylinder

- Centre position : 0,0,1 mm
- Axis : z
- Radius : 1 mm
- Height : 107 mm

3. **Assign material**

➔ the cylinder
➔ Right click on the cylinder
➔ Click on Assign Material
➔ Choose perfect conductorSelect

4. Draw the cylinder(dipole arm). Right click on the **CreateCylinder** to change the dimensions and position of the cylinder.

- Centre position : 0,0,-1 mm
- Axis : Z
- Radius : 1 mm
- Height : -107 mm

5. **Assign material**

➔ Select the cylinder
➔ Right click on the cylinder
➔ Click on Assign Material
➔ Choose perfect conductor

6. Draw the rectangle. Right click on the **CreateRectangle** to change the dimensions and position of the rectangle.

- Position : 0,-1,-1 mm
- Axis : X
- Ysize : 2 mm
- ZSize : 2 mm

7. Select port

➔ Select the rectangle
➔ Right click on the rectangle
➔ Click on Assign Excitation
➔ Choose Lumped Port
➔ Select new line
➔ Draw a line from bottom middle to top middle of the rectangle
➔ Click next
➔ Click finish

8. Draw the Radiation Cylinder

9. Right click on the **CreateCylinder** to change the dimensions and position of the cylinder.

- Centre position : 0,0,-200 mm
- Axis : Z
- Radius : 100 mm
- Height : 400 mm

10. Assign Material

Select cylinder

Right click on cylinder

Click on Assign material

Select Air

Click on Ok

11. Radiation

➔ Select cylinder
➔ Right click on the cylinder
➔ Click on Assign boundary
➔ Click on Radiation
➔ Choose Radiating Only

➔ Click on Ok

12. Solution setup

➔ Go to Analysis
➔ Click on Add solution setup

- Solution frequency : 0.7 GHz
- Number of passes : 10
- Click on Ok

13. **Analysis setup 1**

➔ Go to setup 1
➔ Click on Add frequency sweep

- Sweep type : fast
- Start : 1 MHz
- Stop : 1.5 GHz
- Step size : 0.01

14. Validate

15. Analyze all

Design and Calculations

Given f = 700MHz=0.7GHz

$c = f \lambda$

where $c = 3*10^8$ m/s

$\lambda = c/f = 3*10^8 / 0.7*10^9$

$= 428.5$ mm

Length of the antenna = $\lambda /4$

$= 107.125$ mm

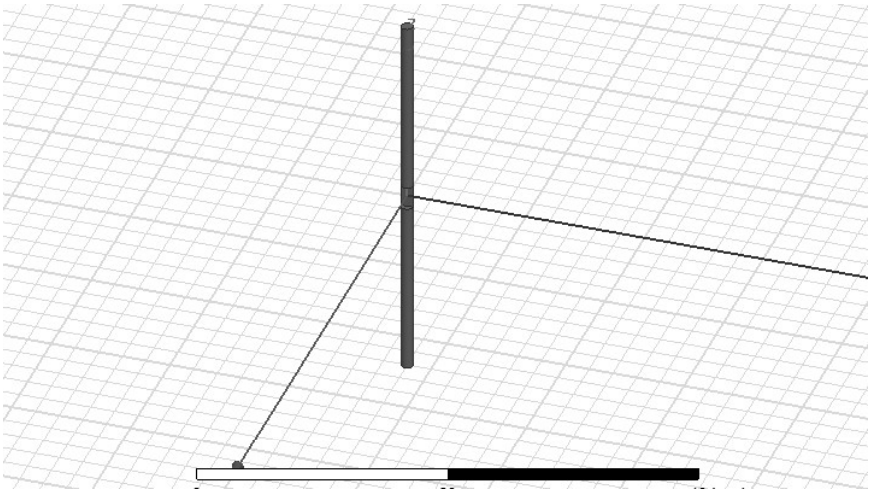
Dimension of radiation cylinder :

Height and Radius of the radiation cylinder measured from the end points of physical dimensions of the antenna $> = 2D^2/\lambda$

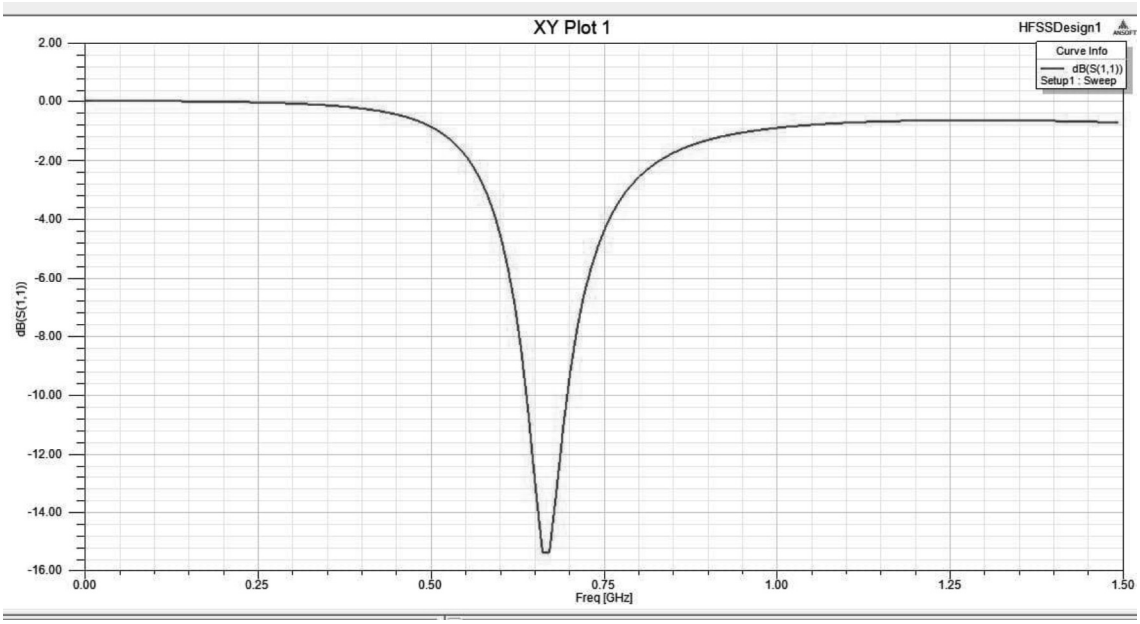
where $D = \lambda/4$

radius, r >= 53.5625 mm

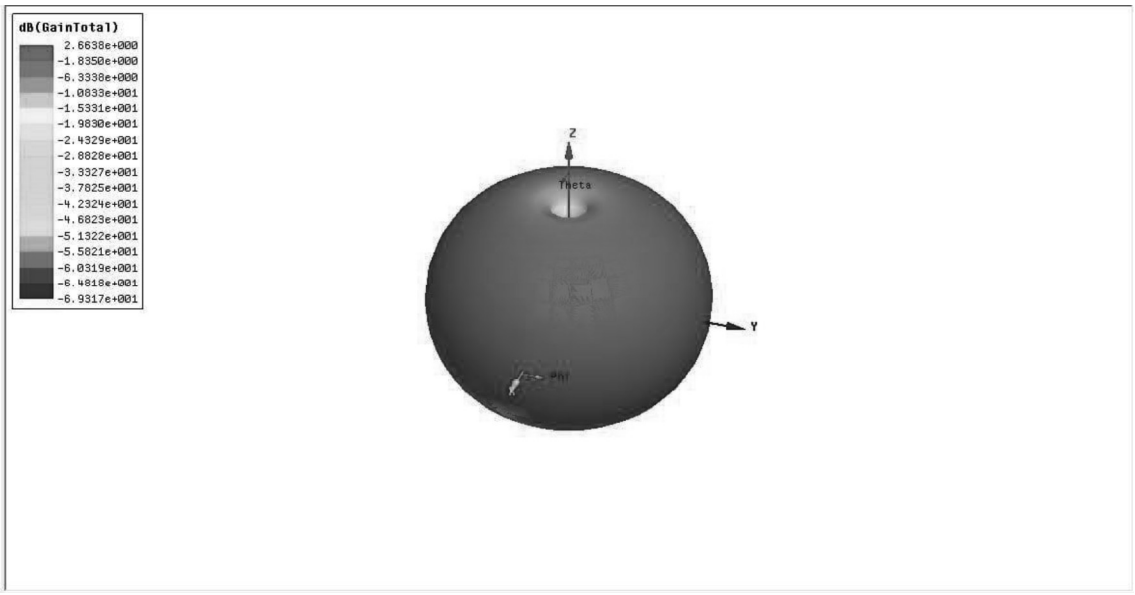
Observation



Return Loss



3D Polar Plot



Result

Designed and simulated a dipole antenna with resonant frequency 0.7 GHz.